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Heat and Mass Transfer Model for a Counter-Flow Moving Packed-Bed Oxidation Reactor/Heat Exchanger

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Particle-based thermochemical energy storage (TCES) through metal oxide redox cycling is advantageous compared to traditional sensible and latent heat storage (SHS and LHS) due to its higher operating temperature and energy density, and the capability for long-duration storage. However, overall system performance also depends on the efficiency of the particle-to-working fluid heat exchangers (HXs). Moving packed-bed particle-to-supercritical CO_2 (sCO_2) HXs have been extensively studied in SHS systems. Integrating the oxidation reactor (OR) for discharging with a particle-to-sCO_2 HX is a natural choice, for which detailed analysis is needed for OR/HX design and operation. In this work, a 2D continuum heat and mass transfer model coupling transport phenomena and reaction kinetics is developed for a shell-and-plate moving-bed OR/HX. For the baseline design, the model predicted ~75% particle bed extent of oxidation at the channel exit, yielding a total heat transfer rate of 16.71 kW for 1.0 m^2 heat transfer area per channel, while the same design with inert particles (SHS only) gives only 4.62 kW. A parametric study was also conducted to evaluate the effects of particle, air, and sCO_2 flowrates, channel height and width, and average particle diameters. It is found that the respective heat transfer rate and sCO_2 outlet temperature can approach ~25 kW and >1000 °C for optimized designs for the OR/HX. The present model will be valuable for further OR/HX design, scale-up, and optimization of operating conditions. [DOI: 10.1115/1.4065040]

Keywords: thermochemical energy storage, moving-bed reactor, supercritical carbon dioxide, particle-based CSP, heat exchanger

1 Introduction

Thermal energy storage (TES) systems based on renewable energy sources are crucial to reducing the dependence on conventional fossil-fuel-based energy generation systems, which are the major sources of air pollution and carbon emissions [1–4]. Sensible, latent, and thermochemical energy storage (TCES) systems are the three major types of TES systems that can overcome the intermittent nature of renewable energy, e.g., wind and solar energy [5]. Compared to traditional electrochemical (e.g., Li-ion) batteries [6], TES could achieve much lower cost and has the potential to become a more cost-effective energy storage method. According to the U.S. Department of Energy [7], molten salts, gas phase, and solid particles are the three major pathways for TES systems. Molten salts are commercially utilized for energy storage, while their low specific heat capacity, low operating temperatures, and component corrosion are the main limitations to their energy conversion efficiency, reliability, and durability. The gas-phase pathway is limited by its high-pressure fatigue, low efficiency due to indirect heat storage, and material corrosion. In contrast, solid particles as energy storage carriers can enable high operating temperatures (e.g., 1000–1500 °C), high storage capacity, high cyclic stability, and low operating costs [8].

Non-reactive particle-based TES systems using sensible heat storage (SHS) are not suitable for long-duration energy storage and have relatively low storage density (1.5–1.8 GJ/m^3) [9], while reactive particle-based TCES systems can store energy for months with higher storage density (up to 3 GJ/m^3) [10]. Different types of reactive particles have been investigated for TCES systems, including salt hydrates, metal hydrides, metal hydroxides, metal carbonates, and metal oxides [5,11–13]. Among these particles, metal oxides have been actively investigated for TCES systems due to their excellent cyclic stability, high energy conversion efficiency, long storage duration, and high operating temperature, which can enable high-efficiency power cycles [14]. Imponenti et al. [15] studied $CaMnO_x$ at 1000 °C for reduction and low O_2 partial pressure (10^{-4} bar) and determined that redox cycles can store up to 700 kJ/kg of chemical energy. Randhir et al. [16–19] established a chemical kinetics model for Mg-Mn-O oxides and developed a packed-bed reactor to demonstrate the charge–discharge cycles between 1000 °C and 1500 °C. Their experimental

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studies suggested that the storage density can be as high as 1070 kJ/kg. Karagiannakis et al. [20] developed a TCES reactor integrated with a heat exchanger (HX) for Co_3O_4/CoO-based materials fabricated in honeycomb structures, in which air was working as both reactants and heat transfer media. To improve cyclic stability, alumina was used in the honeycomb structure to improve the structural integrity without sacrificing the oxidation and reduction rates. Carillo et al. [21] investigated BaO_2/BaO-based TCES redox cycles for high-temperature operation and determined the feasibility for cyclic operation, where good cyclic stability (30 cycles) and high energy density (390 kJ/kg) were observed. Therefore, metal oxide redox cycles are promising for TCES; however, the performance is limited by the availability of materials that can withstand high temperature and scarce indirect HX designs that can address the limitations of direct HXs.

Various reactor-HX assemblies have been studied for different metal oxides in packed, moving, and fluidized bed configurations. Wokon et al. [22] utilized $Mn_{0.75}Fe_{0.25}O_x$ in a packed-bed reactor to study the redox behavior of metal oxides at 800–1040 °C. The reactor wall was made of Inconel 625, which can withstand extremely high temperatures ~1040 °C, and air was used as both the heat transfer fluid (HTF) and reactant. The redox reactions depended on the rate of heat input, heat dissipation, and the partial pressure of O_2. Good cyclic stability was observed, and about 17 cycles were obtained before the material started showing signs of degradation. The thermal reduction of $(Mn_{0.33}Fe_{0.67})_2O_x$ was studied by Wang et al. [23] under high-flux solar irradiation. A detailed 3D computational fluid dynamics (CFD) model coupled with Monte Carlo ray tracing is developed and validated. For the experimental setup, the TCES reactor tube is made of silicon carbide, and a peak flux of 9.5 MW/m^2 was provided to the reactor. The cavity absorption efficiency was 80%, the reaction tube efficiency was 40%, and the solar-to-chemical energy conversion efficiency was 9.3%, which can be improved by utilizing more tubes in the reactor. A honeycomb-based fixed-bed reactor was computationally studied by Singh et al. [24] for Co_3O_4/CoO-based redox reactions, and the numerical model was validated with a 74 kWh_{th} reactor prototype. Parabolic temperature profiles were observed in the reactor along the axis, and relative errors between experimental and numerical energy stored/released were within 9.5%.

Although packed-bed reactor-HX assemblies are widely used in TCES systems, they have limitations of high-pressure drop, non-uniform temperature distribution of heat transfer fluid, and low heat transfer coefficients. Moving-bed reactors integrated with HXs can address these limitations by offering high surface area, high heat transfer coefficients, continuous operation, and uniform heat and mass transfer; hence, are being explored recently in the TCES community [25]. Preisner et al. [26] numerically investigated a counter-current moving-bed reactor for TCES based on Mn-Fe-O with a 1D model, which was validated against the TCES reactor setup in Wokon et al. [22]. Compared to inert particles, a 27% increase in energy density and thermal power was observed, and the oxidation kinetics were determined to be the limiting factor in obtaining higher energy densities. A lab-scale moving-bed reactor with Mg-Mn-O particles undergoing thermochemical reduction in alumina tubes was also demonstrated by Randhir et al. [27], where high-temperature operation (1450 °C) and high thermal-to-chemical reactor efficiencies (~28%) considering heat losses were reported. Huang et al. [28] and Korba et al. [29] developed a 1D and a 2D numerical heat/mass transfer model, respectively, to simulate the experimental work of Randhir et al. [27]. Good validation for local temperature profiles, O_2 concentration, and extent of reaction was obtained using the models, whereby the 2D model was more accurate by capturing the variations in the radial direction.

The reactor-HX assemblies in TCES systems can be classified into direct and indirect types depending on whether the solid particles and working fluid (WF) are in direct contact. Direct reactor-HXs are advantageous in terms of low parasitic energy loss, simple operation strategies, and cost-effectiveness, while the WF may be contaminated by the flowing particles. A moving-bed oxidation reactor (OR) using Mg-Mn-O particles with direct high-temperature (up to 1000 °C) heat extraction to air was demonstrated by Hayes et al. [30], and a detailed computational model was developed by Korba et al. [31] to elucidate the effects of major design and operating parameters on the performance of the direct OR/HX. In contrast, indirect reactor-HXs can avoid WF contamination, utilize various types of WFs irrespective of the reactive media in reaction chambers, and require no post-treatment or pressurization of WFs. The materials to build indirect reactor-HXs should not react with the TCES materials and be able to withstand high operation temperature and pressure. Caccia et al. [32] developed a composite of ZrC and W with a working temperature of 1225 °C and pressure up to 45 MPa for supercritical CO_2 (sCO_2) power cycles. The techno-economic analysis suggested that it is economically viable to use the composite of ZrC and W instead of Ni-based materials. Li et al. [33] designed SiC-based micro-channel HXs that can be operated in air and sCO_2 conditions at 1000–1500 °C, and HX fabrication cost was low. Bader et al. [34] numerically studied a 3 kW moving-bed solar reactor for ceria-based energy storage materials using reticulated porous Al_2O_3 as the reactor wall under high-temperature and stress cycling conditions. Reactor temperatures can be as high as 1500 °C, and the Mohr-Coulomb static factor of safety was >2.5, signifying low stress at elevated operating temperatures. Hence, coupling high-temperature TCES reactors to air or sCO_2 power cycles to enable high-efficiency Brayton or Rankine cycles with WF outlet temperatures >750 °C [7] becomes viable.

Therefore, many efforts have been devoted to developing indirect moving packed-bed heat exchangers within both SHS and TCES categories. For SHS-based indirect HXs coupling with sCO_2 power cycles, extensive previous works can be found, such as Refs. [8,25,35–37], while the literature on TCES-based integrated OR/HXs is scarce, e.g., Refs. [22,24,26]. Hence, the goal of the present study is to fully understand the transport phenomena and oxidation reaction in the OR as well as the heat exchange from the OR to the sCO_2 stream. A 2D continuum heat and mass transfer model for the integrated OR/HX is thus developed. In the present OR/HX design, reduced Mg-Mn-O particles are supplied from the top of the reactor and oxidized by counterflow air from the bottom of the reactor, releasing high-temperature heat in the reaction zone, which is exchanged to the sCO_2 flow in a shell-and-plate HX configuration. Improvement of the OR/HX than the SHS-only HX in terms of overall heat transfer rate and sCO_2 outlet temperature is demonstrated. Furthermore, a comprehensive parametric study was also conducted to evaluate the effects of design and operating conditions, such as the particle capacitance ratio, HX capacitance ratio, particle bed channel aspect ratio, and average particle diameter, on the performance of the OR/HX.

2 Model Development

In this work, a 2D model based on the finite volume method is developed to study the heat transfer enhancement in a counterflow shell-and-plate moving packed-bed OR/HX using ANSYS FLUENT. Section 2.1 describes the OR/HX geometry and the initial estimation of the operating parameters for the HX. Then, the governing equations, constitutive relations, and chemical kinetics are discussed in Secs. 2.2 and 2.3. The temperature-dependent material properties are mentioned in Sec. 2.4, while the boundary conditions are outlined in Sec. 2.5. Finally, Sec. 2.6 describes the numerical methodology employed to couple the heat transfer, fluid flow, and chemical reaction modules.

2.1 OR/HX Configuration and Computational Domain. The schematic for the basic repeating unit of the counterflow moving packed-bed OR integrated with shell-and-plate particle-to-sCO_2 HX is given in Fig. 1. Half the width of the gas-particle channel and sCO_2 channel is considered for computational purposes due to symmetry conditions. It is also assumed that the

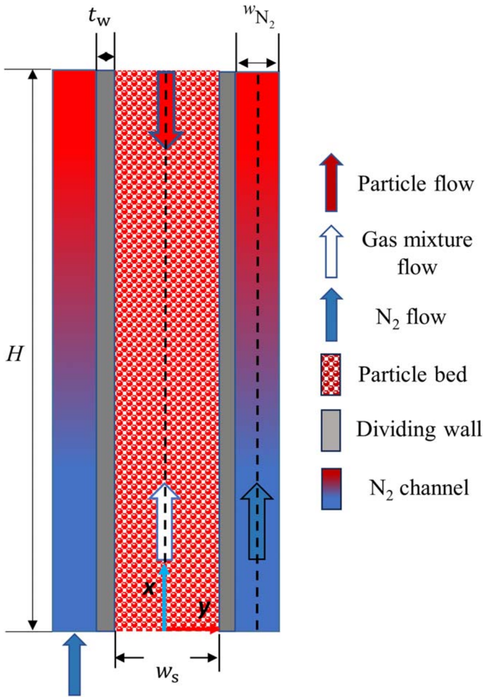

Fig. 1 Schematic of the counterflow gas–solid oxidation reactor integrated with shell-and-plate particle-to-sCO$_2$ heat exchanger (OR/HX) with one channel unit

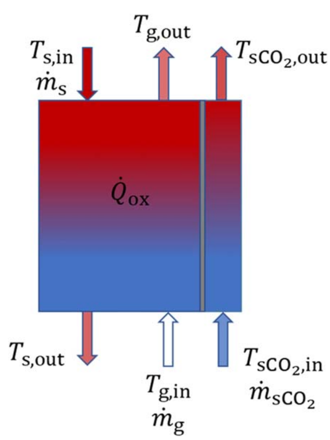

Fig. 2 Schematic of the control volume and steady-state boundary conditions for the TCES-based particle-to-sCO$_2$ HX

channels are deep, and variations in the z-direction are neglected, i.e., a 2D model is used in this work. For the baseline geometry, the channel height (H) is 400 mm, the solid channel width (w_s) is 25.4 mm, the dividing wall (t_w) is 1 mm, and the sCO$_2$ channel width (w_{sCO_2}) is 0.5 mm. Fully reduced particles are fed into the particle channel from the top, and the gas mixture (considering a mixture of O$_2$ and N$_2$ with an O$_2$ mass fraction of 0.231) enters the particle channel from the bottom. The counterflow sCO$_2$ enters the WF channel from the bottom and carries away the heat

Table 1 Steady-state net energy balance equations for baseline case calculations for particle-to-sCO$_2$ HX

$\dot{Q}_s = \dot{Q}_{SH,s} + \dot{Q}_{ox} - \dot{Q}_{SH,g}$	(1*a*)
$\dot{Q}_{SH,s} = \dot{m}_s c_{p,s}(T_{s,out} - T_{s,in})$	(1*b*)
$\dot{Q}_{ox} = \dot{X}_{O_2}\Delta H$	(1*c*)
$\dot{Q}_{SH,g} = \dot{m}_g c_{p,g}(T_{g,out} - T_{g,in})$	(1*d*)
$\dot{Q}_{SH,sCO_2} = \dot{m}_{sCO_2} c_{p,sCO_2}(T_{sCO_2,out} - T_{sCO_2,in})$	(1*e*)

Table 2 Governing differential equations

Gas mixture	Mass balance	
	$\nabla\cdot(\varepsilon\rho_g\vec{v}_g) = \nabla\cdot(D_g\rho_g) + S_{g,m}$	(2)
	Momentum balance	
	$\nabla\cdot(\varepsilon\rho_g\vec{v}_g\vec{v}_g) = -\nabla p_g + \nabla\cdot(\bar{\bar{\tau}}) + \varepsilon\rho_g\vec{g} + \vec{F}_{int} + \vec{F}_{rt}$	(3)
	Species transport	
	$\nabla\cdot(\varepsilon\vec{v}_g\rho_g Y_{O_2}) = \nabla\cdot(D_{O_2}\rho_g Y_{O_2}) + S_{g,m}$	(4)
	Energy balance	
	$\nabla\cdot(\varepsilon\vec{v}_g\rho_g E_g) = \nabla\cdot(\lambda_g\nabla T_g) + h_{gs}A_{gs}(T_s - T_g) + S_{g,e}$	(5)
Particle bed	Species transport	
	$\nabla\cdot((1-\varepsilon)\vec{v}_s\rho_s Y_s) = \nabla\cdot(D_s\rho_s Y_s) + S_{s,s}$	(6)
	Energy balance	
	$\nabla\cdot(\vec{v}_s\rho_s E_s) = \nabla\cdot(\lambda_{s,eff}\nabla T_s) + h_{gs}A_{gs}(T_g - T_s) + S_{s,e}$	(7)
Dividing wall	Energy balance	
	$\nabla\cdot(\lambda_w\nabla T) = 0$	(8)
sCO$_2$	Mass balance	
	$\nabla\cdot(\rho_{sCO_2}\vec{v}_{sCO_2}) = 0$	(9)
	Momentum balance	
	$\nabla\cdot(\rho_{sCO_2}\vec{v}_{sCO_2}\vec{v}_{sCO_2}) = -\nabla p_{sCO_2} + \nabla\cdot(\bar{\bar{\tau}}) + \rho_{sCO_2}\vec{g}$	(10)
	Energy balance	
	$\nabla\cdot(\vec{v}_{sCO_2}\rho_{sCO_2}E_{sCO_2}) = \nabla\cdot(\lambda_{sCO_2}\nabla T)$	(11)

Table 3 Source/sink terms in conservation equations due to chemical reaction

Gas phase/particle bed mass balance and species transport source terms	
$S_{g,m} = (1-\varepsilon)(\rho_{ox} - \rho_{red})\dfrac{d\alpha}{dt}$	(12)
$S_{s,s} = (1-\varepsilon)\rho_s\dfrac{d\alpha}{dt}$	(13)
Gas phase/particle bed energy balance source terms	
$S_{g,e} = (1-\varepsilon)(\rho_{ox} - \rho_{red})\Delta H\dfrac{d\alpha}{dt}$	(14)
$S_{s,e} = -(1-\varepsilon)(\rho_{ox} - \rho_{red})\Delta H\dfrac{d\alpha}{dt}$	(15)

Table 4 Constitutive heat, mass, and momentum transport equations

Gas mixture/sCO$_2$ stress–strain tensor [43]

$$\bar{\bar{\tau}} = \mu_{g/sCO_2}(\nabla \vec{v}_{g/sCO_2} + \nabla \vec{v}^{T}_{g/sCO_2}) + \left(\lambda_{g/sCO_2} - \frac{2}{3}\mu_{g/sCO_2}\right)\nabla \cdot \vec{v}_{g/sCO_2}\bar{\bar{I}} \tag{16}$$

Particle bed–gas mixture interfacial forces (Gidaspow model) [44]

$$\vec{F}_{int} = K_{sg}(\vec{v}_s - \vec{v}_g) \tag{17}$$

Gas mixture–particle bed exchange coefficient

$$K_{sg} = \begin{cases} 150\frac{(1-\varepsilon)^2}{\varepsilon}\frac{\mu_g}{d_s^2} + 1.75(1-\varepsilon)\frac{\rho_g}{d_s}|\vec{v}_g - \vec{v}_s| & (0 \le \varepsilon \le 0.8) \\ \frac{3}{4}C_D\frac{\rho_g(1-\varepsilon)\varepsilon^{-1.65}}{d_s}|\vec{v}_g - \vec{v}_s| & (0.8 < \varepsilon \le 1) \end{cases} \tag{18}$$

$$C_D = \frac{24}{\varepsilon Re_s}[1 + 0.15(\varepsilon Re_s)^{0.687}] \tag{19}$$

Interfacial momentum transport due to reaction

$$\vec{F}_{rt} = S_{g,m}(\vec{v}_s - \vec{v}_g) \tag{20}$$

Gas phase–particle bed heat exchange coefficient [45]

$$h_{gs} = \frac{\lambda_g Nu}{d_s} \tag{21}$$

$$Nu = (7 - 10\varepsilon + 5\varepsilon^2)(1 + 0.7Re_s^{0.2}Pr^{1/3}) + (1.33 - 2.4\varepsilon + 1.2\varepsilon^2)Re_s^{0.7}Pr^{1/3} \tag{22}$$

Interfacial area

$$A_{gs} = \frac{6\varepsilon(1-\varepsilon)}{d_s} \tag{23}$$

Effective particle phase thermal conductivity [26,28,29]

$$\lambda_{s,eff} = \lambda_{bulk,eff} - \varepsilon_{bed}\lambda_g \tag{24}$$

Reynolds number

$$Re_s = \frac{\varepsilon \rho_g d_s |\vec{v}_g - \vec{v}_s|}{\mu_g} \tag{25}$$

Prandtl number

$$Pr = \frac{c_{p,g}\mu_g}{k_g} \tag{26}$$

Near-wall thermal resistance [46]

$$R_{nw} = \frac{d_s}{2\lambda_{s,nw}} \tag{27}$$

Near-wall particle bed effective thermal conductivity [47,48]

$$k^{nw}_{bed,eff} = k_{air}\left(\varepsilon_{nw} + \frac{1-\varepsilon_{nw}}{2\Phi_{nw} + \frac{2k_{air}}{3k_s}}\right) \tag{28}$$

Near-wall voidage [46]

$$\varepsilon_{nw} = 1 - \frac{(1-\varepsilon_{bed})\left(0.7293 + 0.5139\frac{d_s}{2a}\right)}{\left(1 + \frac{d_s}{2a}\right)} \tag{29}$$

from the particle channel. For the sensible heat-only HX, the gas flow in the particle channel is not present.

An initial steady-state energy balance analysis for the OR/HX control volume (CV) is conducted to estimate the baseline design and operating parameters based on the schematic in Fig. 2. The inlet temperatures are chosen at $T_{s,in} = 800$ °C, $T_{sCO_2,in} = 550$ °C, and $T_{g,in} = 550$ °C. The particle outlet temperature ($T_{s,out}$) is estimated to be ∼550 °C, the gas mixture outlet temperature ($T_{g,out}$) is estimated to be ∼800 °C to match the particle inlet temperature, and the sCO$_2$ outlet temperature ($T_{sCO_2,out}$) is estimated at ∼700 °C based on typical WF outlet temperatures for sensible HXs [38,39]. The subscripts "g," "s," and "sCO$_2$" denote the gas mixture, solid particle bed, and sCO$_2$ gas stream, respectively. In addition, a 7-kW power output for the OR/HX (half channel) is selected based on previous sensible HX designs that operated in the range of 4.5–10 kW [39–41]. The thermochemical power ($\dot{Q}_{ox}$) released from the oxidation of the Mg-Mn-O particles is estimated to be ∼6 kW, for which 0.015 mol/s of O$_2$ is required, corresponding to 2.5 g/s of airflow. To ensure that enough O$_2$ is present in the particle channel, a gas flowrate $\dot{m}_g = 3$ g/s is chosen, which is below the minimum fluidization velocity of the particles [42]. From Ref. [16], a particle bed flowrate $\dot{m}_s = 10\,\mathrm{g/s}$ is required to absorb 0.015 mol/s of O_2. Therefore, as summarized in Table 1, the total heat transfer rate on the particle side ($\dot{Q}_s$) is given by Eq. (1*a*) and sCO_2 side ($\dot{Q}_{SH,sCO_2}$) is given by Eq. (1e), where the subscripts SH and OX denote the sensible heat and oxidation heat transfer, respectively. From Eq. (1*a*), $\dot{Q}_s$ is evaluated to be 7 kW, so based on the energy balance in the CV, $\dot{Q}_s = \dot{Q}_{SH,sCO_2}$. Hence, the mass flowrate of sCO_2 is 38 g/s equating Eqs. (1*a*) and (1*e*) $\left(\dot{m}_{sCO_2} = \frac{\dot{Q}_{SH,sCO_2}}{c_{p,sCO_2}(T_{sCO_2,out} - T_{sCO_2,in})}\right)$. It should be noted that here, all the mass flowrates and inlet and outlet temperatures are assumed to be constant for the steady-state energy balance analysis only. The actual outlet temperatures for a particular case will be determined based on the physical model and detailed simulations, as shown in Sec. 4.

2.2 Governing and Constitutive Equations. A steady-state 2D CFD model is developed for a counterflow moving-bed OR/HX. The governing equations for the reactive/inert moving packed-bed, interstitial gas mixture through the reactive particle bed, dividing wall, and sCO_2 flow are summarized in Table 2. The subscript "w" denotes the dividing wall. The solution variables for Eqs. (2)–(11) are density (ρ), velocity ($\vec{v}$), pressure (p), temperature (T), and mass fractions of O_2 in the gas mixture ($Y_{O_2} = m_{O_2}/m_{total}$) and reduced particle mass in the total solid mass ($Y_s = m_{red}/m_{total}$). In those equations, ε is the porosity of the particle bed, D is the diffusivity, λ is the thermal conductivity, E is the internal energy, h is the interstitial heat transfer coefficient, and A is the specific surface area. For the gas mixture, the O_2 mass fraction is tracked using Eq. (4), whereas for the particle bed, the mass fraction of the reduced particles is tracked using Eq. (6). The particle bed momentum equation is not solved since a constant velocity is used. The mass, momentum, species, and energy source/sink terms due to chemical reaction are mentioned in Tables 3 and 4. Here, $S_{g,m}$ is the mass generation/consumption source term for the gas mixture, $S_{g,e}$ is the energy source/sink term for the gas mixture, $S_{s,s}$ is the reduced particle bed mass generation/consumption source term, $S_{s,e}$ is the energy source/sink term for the particle bed, and $\vec{F}_{rt}$ is the interfacial momentum transfer due to reaction.

The constitutive equations for heat, mass, and momentum transport are given in Table 4. The gas mixture/sCO_2 stress–strain tensor follows the convection of Bird [43]. The interfacial forces between the moving particle bed and gas mixture flow stream are defined by the Gidaspow model [44]. The widely adopted Gunn [45] correlation is used to define the heat transfer coefficient between the particle bed and gas mixture, and the specific surface area is given by Eq. (22) and is valid for $0.35 < \varepsilon < 1$. The particle bed's effective thermal conductivity can be estimated as Refs. [26,28,29], and the near-wall particle bed thermal conductivity is accounted for by Refs. [46–48].

2.3 Reaction Kinetics. The reactive material magnesium manganese oxide with a molar ratio MgO/MnO = 1:1 is used in this work as the particle bed, and the thermochemical reversible reaction is given by Refs. [16,17]

$$\mathrm{Mg_xMnO_{1+x+y_1}} + \Delta H \rightleftharpoons \mathrm{Mg_xMnO_{1+x+y_2}} + \frac{y_1 - y_2}{2}\mathrm{O_2} \tag{30}$$

where $\mathrm{Mg_xMnO_{1+x+y_2}}$ is the reduced material, and $\mathrm{Mg_xMnO_{1+x+y_1}}$ is the oxidized material. The oxidation reaction kinetics of Mg-Mn-O are adopted from Bo et al. [19] and summarized in Table 5, and the kinetic model parameters are mentioned in Refs. [28,49]. The reaction rate has local temperature, partial pressure of oxygen, and extent of conversion dependency and is accounted for in this model. For α, the fully oxidized and fully reduced states correspond to 1 and 0, respectively.

Table 5 Reaction kinetics model

Extent of conversion

$$\alpha = \frac{m - m_{red}}{m_{ox} - m_{red}} \tag{31}$$

Reaction rate

$$\frac{d\alpha}{dt} = A_0 e^{-\frac{Ea}{RT}}\left(\frac{p_{O_2}}{p_{ref}}\right)^n \frac{f(\alpha)}{T}(\mu_{O_2,s} - \mu_{O_2,g}) \tag{32}$$

$$f(\alpha) = \begin{cases} \alpha, & \sigma_{O_2,s} > \sigma_{O_2,g} \\ 1-\alpha, & \sigma_{O_2,s} < \sigma_{O_2,g} \end{cases} \tag{33}$$

Molar Gibbs free energy at which the spinel to monoxide phase transformation occurs

$$\mu_{O_2,g} = \mu^{o}_{O_2} + RT\ln\left(\frac{p_{O_2}}{p_{ref}}\right) \tag{34}$$

Molar Gibbs free energy

$$\mu_{O_2,s} = \sum_{i=0}^{6} a_i X^i - \lambda T\alpha \tag{35}$$

$$X = \ln(\alpha + \alpha_0)\left[1 - \frac{H(\gamma)}{T}\right] + \frac{H(\gamma)\beta\gamma}{T} \tag{36}$$

2.4 Material Properties. The thermophysical properties of particle bed, gas mixture, and sCO_2 are considered temperature dependent in this work between the operating temperatures of 550 °C and 1200 °C and are summarized in Table 6. The particle bed properties of Mg-Mn-O are obtained from Refs. [28,29], the gas mixture properties from Refs. [28,29], the sCO_2 properties are evaluated at 25 MPa [51], and the dividing wall is assumed to have constant thermophysical properties and is adopted from Torrijios et al. [52]. As the pressure drop in the moving bed is much lower than that in a packed bed, the pressure effect on thermophysical properties was not considered, and all properties in the particle channel are evaluated at atmospheric pressure.

2.5 Boundary Conditions. The boundary conditions of the OR/HX are given in Table 7, where a constant inlet mass flowrate/temperature boundary condition is used for the particle bed, gas mixture, and sCO_2. Pressure outlet boundary condition is used for the gas mixture (101,325 Pa) and sCO_2 (25 MPa), and a plug flow assumption is employed for the particle bed flow. Fully developed boundary conditions are adopted for temperature and species at the particle bed outlet/inlet and sCO_2 outlet. Fully reduced particles enter the reactor ($\alpha = 0$), and the gas mixture is considered as a mixture of O_2 and N_2 with $Y_{O_2,in} = 0.231$. Due to geometric symmetry, only half channel of the particle bed and sCO_2 are simulated, and the dividing wall is insulated at the two ends. No-slip boundary conditions are imposed on solid walls, and standard conjugate conditions are used at the interfaces of dissimilar materials. The dimensions of the OR/HX and particle diameter are given in Eqs. (40) and (41).

2.6 Numerical Methodology. The mass, momentum, species, and energy conservation equations are solved using the finite volume method in ANSYS FLUENT 22.2. The convective terms are discretized using a second-order upwind scheme, and the gas mixture and sCO_2 flows are assumed laminar. A pressure-based solver is used, and the phase-coupled SIMPLE algorithm couples the pressure and velocity terms. The source/sink terms (Table 3), reaction kinetics (Table 5), and temperature-dependent thermophysical

Table 6 Material thermophysical properties

Property	Value	Unit	Reference
λ_g	$1.01805\times10^{-20}T^6 - 1.71098\times10^{-17}T^5 - 2.26255\times10^{-14}T^4 + 8.53554\times10^{-11}T^3 - 1.03690\times10^{-7}T^2 + 1.14337\times10^{-4}T - 1.08670\times10^{-3}$	W/m/K	[29]
$\lambda_{s,eff}$	$2.5760\times10^{-17}T^6 - 1.0400\times10^{-13}T^5 + 1.7301\times10^{-10}T^4 - 1.5273\times10^{-7}T^3 + 8.0236\times10^{-5}T^2 - 3.1566\times10^{-2}T + 12.9662$	W/m/K	[50]
λ_{sCO_2}	$3.0\times10^{-9}T^2+7.0\times10^{-5}T+0.0082$	W/m/K	[51]
λ_w	23	W/m/K	[40]
$c_{p,g}$	$\begin{cases}\begin{bmatrix}-1.573588\times10^{-17}T^7+6.540196\times10^{-14}T^6-1.111097\times10^{-10}T^5+\\ 9.92857\times10^{-8}T^4-5.034909\times10^{-5}T^3+0.01485511T^2\\ -2.368819T+1161.482\end{bmatrix} & 100\,\mathrm{K}\le T_g\le 1000\,\mathrm{K}\\ \begin{bmatrix}1.112335\times10^{-19}T^7-1.565553\times10^{-15}T^6+9.237533\times10^{-12}T^5-\\ 2.936679\times10^{-8}T^4+5.421615\times10^{-5}T^3-0.0581276T^2\\ +33.70605T-7069.814\end{bmatrix} & 1000\,\mathrm{K}< T_g\le 3000\mathrm{K}\end{cases}$	J/kg/K	[29]
$c_{p,s}$	$-2.503\times10^{-16}T^6 + 1.4394\times10^{-12}T^5 - 3.228\times10^{-9}T^4 + 3.6473\times10^{-6}T^3 - 2.3635\times10^{-3}T^2 + 1.0435T + 676.24$	J/kg/K	[29]
c_{p,sCO_2}	$3.0\times10^{-5}T^2+8.79\times10^{-2}T+1160.70$	J/kg/K	[51]
$c_{p,w}$	871	J/kg/K	[52]
ρ_s	4525	kg/m^3	[29]
ρ_{sCO_2}	$2.0\times10^{-4}T^2-4.7610^{-1}T+429.82$	kg/m^3	[51]
ρ_w	2719	kg/m^3	[52]
μ_g	$\dfrac{1.458\times10^{-6}T^{3/2}}{T+110.4}$	kg/m/s	[29]
μ_{sCO_2}	$-2.0\times10^{-12}T^2-3.0\times10^{-8}T+2.0\times10^{-5}$	kg/m/s	[51]

Table 7 Boundary conditions and baseline design parameters

Particle bed, gas mixture, and sCO_2 mass flowrates

$$\dot{m}_{s,in}=10\,\mathrm{g/s},\ \dot{m}_{g,in}=3\,\mathrm{g/s},\ \dot{m}_{sCO_2,in}=38\,\mathrm{g/s} \quad (37)$$

Particle bed, gas mixture, and sCO_2 inlet temperatures

$$T_{s,in}=800\,^\circ\mathrm{C},\ T_{g,in}=550\,^\circ\mathrm{C},\ T_{sCO_2,in}=550\,^\circ\mathrm{C} \quad (38)$$

Reduced particle bed inlet and O_2 inlet mass fractions

$$Y_{s,in}=0,\ Y_{O_2,in}=0.231 \quad (39)$$

Particle bed, wall and sCO_2 width

$$w_s=25.40\,\mathrm{mm},\ w_w=1\,\mathrm{mm},\ w_{sCO_2}=0.5\,\mathrm{mm} \quad (40)$$

HX height and particle diameter

$$H=400\,\mathrm{mm},\ d_s=1\,\mathrm{mm} \quad (41)$$

properties (Table 6) are implemented through user-defined functions (UDFs) in the FLUENT framework. The solution is assumed to have reached steady-state/convergence when the residuals fall below 10^{-6} for mass and momentum equations and 10^{-12} for the species and energy equations.

3 Reduced Model Verification

In this section, model verification for the SHS-only particle-to-sCO_2 HX is conducted. For mesh independence, the average particle bed-to-wall heat transfer coefficient, $\bar{h}_{s,w}$, from the present model is compared with that from Albrecht and Ho [39] for the baseline case in Table 8 at different mesh sizes, where the relative error (R.E.) is defined as

$$\mathrm{R.E.}=\left|\frac{\bar{h}_{s,w}-\bar{h}_{s,w,ref}}{\bar{h}_{s,w,ref}}\right|\times100\% \quad (42)$$

Table 8 Mesh independence study for average particle bed-to-wall heat transfer coefficient comparison with Albrecht and Ho [39]

Mesh size (nodes)	$\bar{h}_{s,w}$ (W/m^2K)	Relative error (%)
12,052	183.5	0.83
31,567	182.81	0.45
78,082	182.34	0.19
125,400	182.29	0.16

Note: $\bar{h}_{s,w,ref} = 182\,W/m^2K$

The R.E. decreases with an increase in mesh size from 12,052 nodes to 125,400 nodes and asymptotes around 78,082 nodes. Hence, a mesh size of 78,082 is chosen, which sufficiently captures the flow and heat transfer in the HX for the rest of the work.

The numerical data from Albrecht and Ho [39] for a counterflow shell-and-plate moving packed-bed heat exchanger (MPBHX) is used to verify the SHS HX model in this work. The nominal parameters are adopted from the baseline case of Albrecht and Ho [39], and the temperature profiles for the particle bed, dividing wall, and sCO_2 along the streamwise direction and $\bar{h}_{s,w}$ variation are compared. It can be observed in Fig. 3 that the temperature profiles all show reasonable agreement. The $\bar{h}_{s,w}$ variations in Fig. 4 for different particle channel width and particle diameter values also show good agreement, verifying our SHS HX model.

4 Results and Discussion

The verified sensible heat-only HX model is coupled with the previously validated reaction kinetics model [16,19,28,29] to simulate the OR/HX baseline case in comparison with the counterpart with SHS only (i.e., when the reaction is turned off). Afterward, the effects of key design and operating parameters on the OR/HX performance are elucidated through a parametric study.

4.1 Comparison of TCES-Based OR/HX With Sensible Heat Storage–Based HX.

The steady-state simulation results for the baseline OR/HX design (key parameters summarized in Table 7) are shown in this section. The 2D temperature contours of the particle bed are shown in Fig. 5(*a*) for sensible heat HX and Fig. 5(*b*) for the OR/HX. It can be observed that due to particle bed oxidation, there is a high-temperature zone (∼1100 °C) developed in the OR/HX, which is significantly higher than the particle inlet temperature. In the sensible heat HX, an average particle bed outlet temperature of 603 °C is obtained compared to 695 °C in the OR/HX.

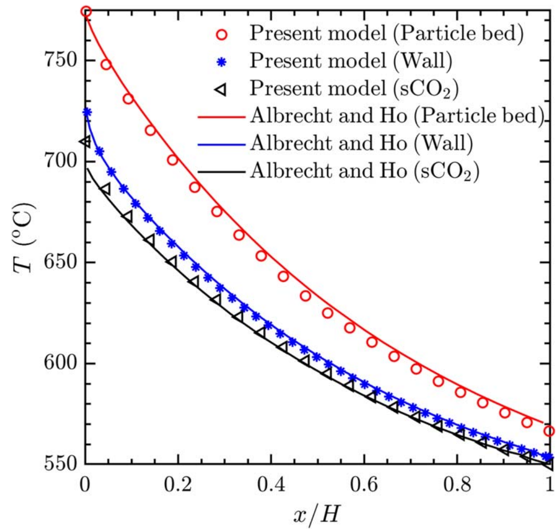

Fig. 3 Temperature profile comparison of particle bed, dividing wall, and sCO_2 along the axial direction with numerical data from Albrecht and Ho [39] ($H = 1$ m, $w_s = 6\times10^{-3}$ m, $w_{sCO_2} = 5\times10^{-4}$ m, $t_w = 1\times10^{-3}$ m, $\dot{m}_s = 23.8$ g/s, $\dot{m}_{sCO_2} = 31.3$ g/s, $T_{s,in} = 775\,°C$, $T_{sCO_2,in} = 550\,°C$, and $d_s = 250\times10^{-6}$ m)

Figure 6 shows the contours of the O_2 mass fraction and extent of reaction (α) in the particle channel. The high-temperature zone in the OR/HX aids oxidation of the incoming particles, yielding high conversion $\alpha \sim 0.7$ at the exit of the particle channel ($x/H = 0$), as seen in Fig. 6(*b*).

A direct comparison of the temperature distributions can also be made by plotting the streamwise averaged temperature profiles of the particle bed, gas mixture, and sCO_2, as shown in Fig. 7. The results are consistent with the contours shown in Fig. 5, with much higher temperatures observed in the OR/HX due to exothermic reaction, and a peak average temperature close to 1100 °C is noted near $x/H = 0.65$. The resulting sCO_2 outlet temperatures are 595 °C and 693 °C for the sensible heat HX and OR/HX, respectively, as shown in Fig. 7(*b*). The total heat transfer rate is calculated to be $\dot{Q}_{OR/HX} = 16.71$ kW for a 1.0 m^2 heat transfer area, which is much higher than the capacity of the sensible heat HX ($\dot{Q}_{SH} = 4.62\,kW$).

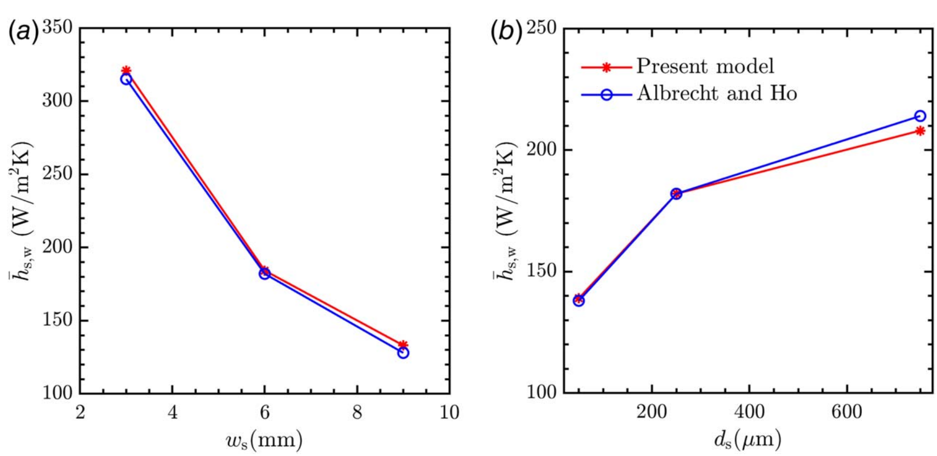

Fig. 4 Average particle bed-to-wall heat transfer coefficient, $\bar{h}_{s,w}$, comparison with Albrecht and Ho [39] for variation in (*a*) particle bed channel width and (*b*) particle diameter ($H = 1$ m, $w_{sCO_2} = 5\times10^{-4}$ m, $t_w = 1\times10^{-3}$ m, $\dot{m}_s = 23.8$ g/s, $\dot{m}_{sCO_2} = 31.3$ g/s, $T_{s,in} = 775\,°C$, and $T_{sCO_2,in} = 550°C$)

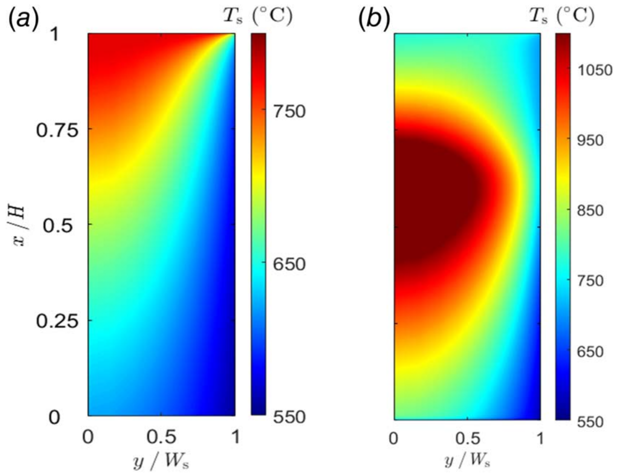

Fig. 5 Particle bed temperature contours for baseline cases of (*a*) sensible heat HX and (*b*) integrated OR/HX

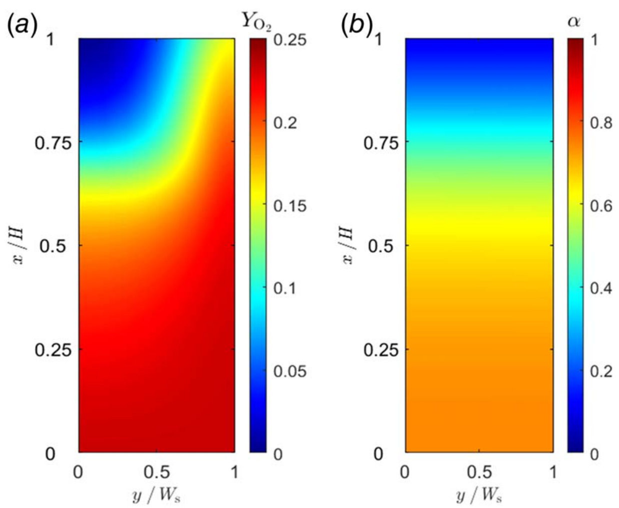

Fig. 6 Contours of (*a*) O_2 mass fraction (Y_{O_2}) and (*b*) extent of reaction (α) for the baseline OR/HX case

From this comparison, it can be observed that for the baseline parameters considered here, including particle bed inlet temperature at 800 °C and particle channel width of 25.40 mm, the sensible heat HX does not perform comparably from the heat exchange capacity perspective.

To ensure optimal operating conditions are utilized while comparing OR/HXs and sensible heat HXs, it is pertinent to use narrow particle channels ($w_s = 3$ mm) as recommended by Albrecht and Ho [39,40], Fang et al. [41], and Yin et al. [53]. Hence, for sensible heat HX, the thermophysical properties, operating conditions, and geometric parameters are adopted from Albrecht and Ho [39], and simulations are carried out for 800 °C < $T_{s,in}$ < 1000 °C. It is to be noted that for the OR/HX, the $T_{s,in}$ is at 800 °C, but for sensible heat, HXs $T_{s,in} > 800$ °C are explored to investigate typical $\dot{Q}$ values that can be obtained with higher energy input via the HTF. Typically utilized sensible heat-based particles like CARBO Accucast ID [25,54], CARBOBEAD CP [55,56], and particles used in the work of Albrecht and Ho [39] are used in the simulation with an average particle diameter of 250 μm and respective thermophysical properties.

Table 9 highlights the heat transfer performance of the sensible heat HX for the three particle types and $T_{s,in}$ values. Higher $\dot{Q}$ is obtained with higher $T_{s,in}$ as expected, with $\dot{Q}$ values ranging between 10 kW and 20 kW for 1 m^2 of heat transfer area, which lies in a similar $\dot{Q}$ output for OR/HXs, with the latter exhibiting slightly higher $\dot{Q}$ values. Furthermore, an in-depth comparison of heat transfer performance is also made with Albrecht and Ho's [39] work on sensible heat HXs with parametric variations in Table 10, where the reference values are compared with the present OR/HX $\dot{Q}$ output. An apparent enhancement in $\dot{Q}$ is observed for the OR/HX compared to sensible heat HX. Although OR/HXs can significantly improve WF outlet temperature and $\dot{Q}$, a system-level efficiency and techno-economic analysis accounting for material costs for high-temperature materials is needed for a complete evaluation.

4.2 Parametric Study of OR/HX Design and Operating Conditions. The effect of key operating parameters on the heat transfer performance of the OR/HX is studied in this section. For the parametric study, baseline parameters are adopted from Table 7 and varied to study the effect of variance in the parameter of interest on the heat and mass transfer performance of the oxidation reactor/heat exchanger. Table 10 outlines the major parameters considered, their definition, and range, where C_{bed} is the particle bed capacitance ratio, C_{HX} is the HX capacitance ratio, AR_s is the aspect ratio of the particle channel, and d_s is the average particle diameter.

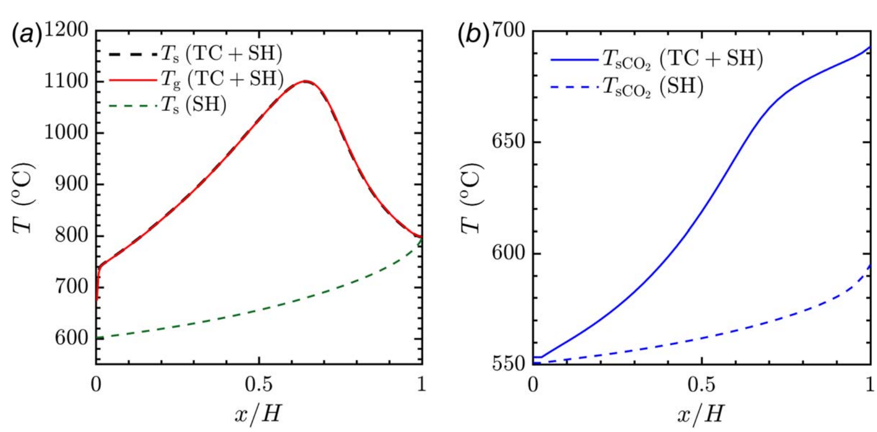

Fig. 7 Streamwise averaged temperature profiles for baseline case (TC-thermochemical and SH-sensible) for (*a*) particle channel (particle bed and gas mixture) and (*b*) sCO_2 channel ($H = 0.4$ m, $w_s = 25.4 \times 10^{-3}$ m, $w_{sCO_2} = 5 \times 10^{-4}$ m, $t_w = 1 \times 10^{-3}$ m, $\dot{m}_s = 10$ g/s, $\dot{m}_{g,in} = 3$ g/s, $\dot{m}_{sCO_2} = 38$ g/s, $T_{s,in} = 800$ °C, $T_{g,in} = 550$ °C, $T_{sCO_2,in} = 550$ °C, and $d_s = 1 \times 10^{-3}$ m)

Table 9 Heat transfer rate for 1 m^2 heat transfer area of sensible heat HX with narrow particle channels (3 mm) at different particle inlet temperature $T_{s,in}$

	$\dot{Q}$(kW)		
$T_{s,in}$	CARBO accucast ID	CARBOBEAD CP	Albrecht and Ho [39]
800 °C	10.76	10.95	11.01
900 °C	15.07	15.33	15.42
1000 °C	19.38	19.71	19.82

Table 10 Heat transfer rate summary for 1 m^2 heat transfer area of sensible heat HX compared with OR/HX

Reference	Parameter		$\dot{Q}$ (kW)
Albrecht and Ho [39]	w_s (mm)	3	10.00
		6	5.86
		9	4.09
	d_s (μm)	50	6.63
		250	5.86
		750	4.50
Present work	$C_{bed} = \frac{\dot{m}_g c_{p,g}}{\dot{m}_s c_{p,s}}$	0.2	8.58
		0.3	12.04
		0.4	17.03
		0.5	17.66
		0.6	17.73
	$C_{HX} = \frac{\dot{m}_{sCO_2} c_{p,sCO_2}}{\dot{m}_s c_{p,s}}$	1.5	17.90
		3	16.91
		4.5	16.56
		6	16.42
		8.5	16.02
	$AR_s = \frac{H}{w_s}$	16	24.54
		24	20.62
		32	16.45
		40	12.24
		48	7.76
	d_s (mm)	0.5	18.97
		0.75	17.50
		1	16.48
		2	11.26

4.2.1 Effect of Particle Bed Capacitance Ratio. The effects of C_{bed} are studied for the OR/HX in this section, where $\dot{m}_s$ is kept constant, and $\dot{m}_g$ is varied to alter C_{bed} within the range of 0.2–0.6. The particle bed temperature profiles are shown in Fig. 8, where the shift in the high-temperature zone can be observed from $x/H = 0$ to $x/H = 1$ as C_{bed} increases from 0.2 to 0.6. For $C_{bed} = 0.2$, the high-temperature zone develops around $x/H = 0$–0.1 and starts moving up to $x/H = 0.25$ for $C_{bed} = 0.3$, eventually settling around $x/H = 0.5$–$x/H = 0.85$ for higher C_{bed} values. This stark difference in the high-temperature zone locations is due to the increase in the gas mixture velocity, which shifts the reaction zone upward and enhances the O_2 in the particle bed to facilitate the oxidation reaction higher up in the particle bed.

The streamwise averaged particle bed temperature profiles in Fig. 9(*a*) clearly outlines the shift in the high-temperature zone upward with an increase in C_{bed}, whereby significantly higher temperatures are observed in higher C_{bed} values. This increase in particle bed temperature can be attributed to enhanced oxidation reaction due to ample O_2 being present in the particle bed, which is supported by the O_2 mass fraction and reaction extent profiles in Figs. 9(*c*) and 9(*d*), respectively. For the sCO_2 streamwise averaged temperature profiles, a higher outlet sCO_2 temperature is observed for higher C_p due to the upward shift of the high-temperature zone in Fig. 9(*b*). For $C_{bed} = 0.2$ and 0.3, due to the high-temperature zone near $x/H = 0$–0.25, there is an increase in the initial sCO_2 temperature, but the growth is not sustained throughout the channel. The total heat transfer rate values for variation in C_{bed} are presented in Table 10, where an increase in $\dot{Q}$ is observed with an increase in C_p from 8.58 kW to 17.73 kW. This increase in $\dot{Q}$ is due to a higher oxidation reaction, which enables greater heat transfer, although beyond $C_{bed} = 0.4$, the increase becomes negligible because of the excess gas flow advecting heat out of the particle bed.

4.2.2 Effect of HX Capacitance Ratio. The effects of C_{HX} (between 0.5 and 8.5) on the OR/HX performance are investigated here, where $\dot{m}_s$ is kept constant and $\dot{m}_{sCO_2}$ is modified. The impact on the particle bed temperature profiles is shown in Fig. 10, where for $C_{HX} = 0.5$, a sizeable high-temperature zone is observed, which starts decreasing and shifting downward as C_{HX} increases. This is due to a higher amount of heat being extracted from the particle bed via the sCO_2 channel. High temperatures near the dividing wall, as in the case of $C_{HX} = 0.5$, may cause significant thermomechanical stresses compared to the higher C_{HX} cases and can decrease the working life of the dividing wall.

The streamwise averaged temperature profiles of the particle bed are shown in Fig. 11(*a*) for varying C_{HX}, and a high-temperature zone ∼1300 °C is observed for $C_{HX} = 0.5$. All the high-temperature zones are located between $0.3 < x/H < 0.9$ and directly influence the sCO_2 outlet temperatures, as seen in Fig. 11(*b*). The sCO_2 outlet temperature decreases as C_{HX} increases, where a peak outlet

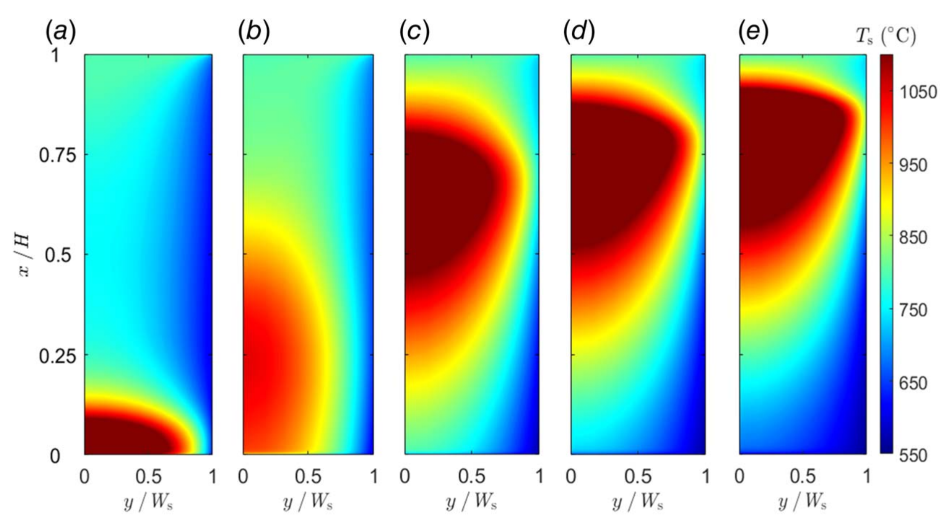

Fig. 8 Particle bed temperature contours for (*a*) $C_{bed} = 0.2$, (*b*) $C_{bed} = 0.3$, (*c*) $C_{bed} = 0.4$, (*d*) $C_{bed} = 0.5$, and (*e*) $C_{bed} = 0.6$

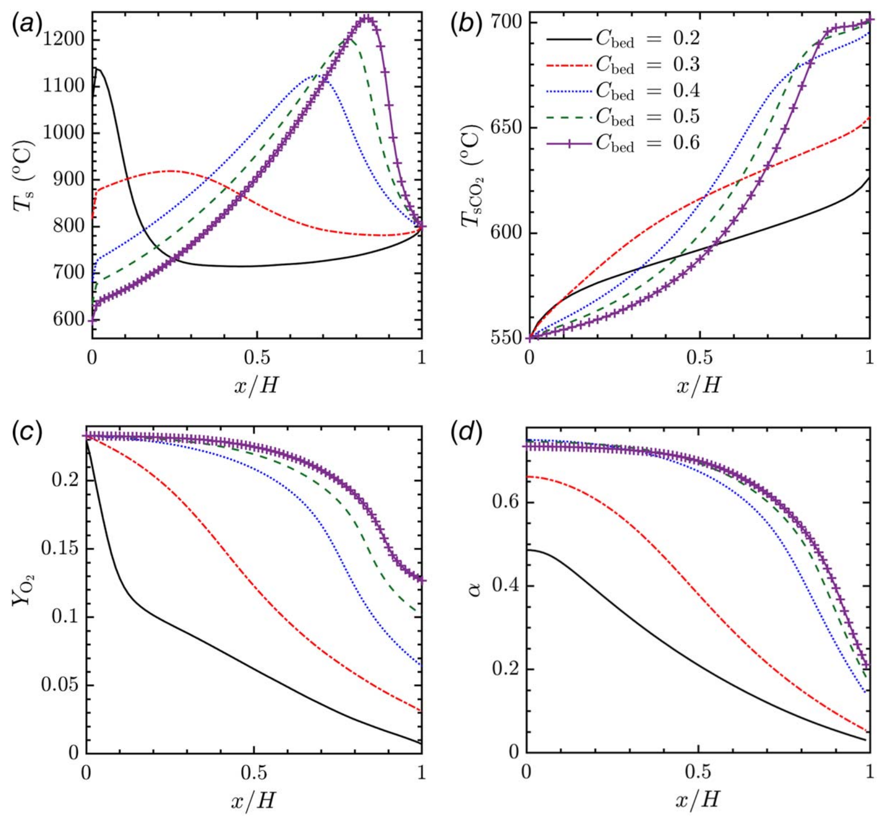

Fig. 9 (*a*) Particle bed temperature, (*b*) sCO_2 temperature, (*c*) O_2 mass fraction, and (*d*) extent of reaction profiles along the particle flow direction for different C_{bed} ($H = 0.4$ m, $w_s = 25.4 \times 10^{-3}$ m, $w_{sCO_2} = 5 \times 10^{-4}$ m, $t_w = 1 \times 10^{-3}$ m, $\dot{m}_s = 10$ g/s, $\dot{m}_{sCO_2} = 38$ g/s, $T_{s,in} = 800\,^\circ$C, $T_{g,in} = 550\,^\circ$C, $T_{sCO_2,in} = 550\,^\circ$C, and $d_s = 1 \times 10^{-3}$ m)

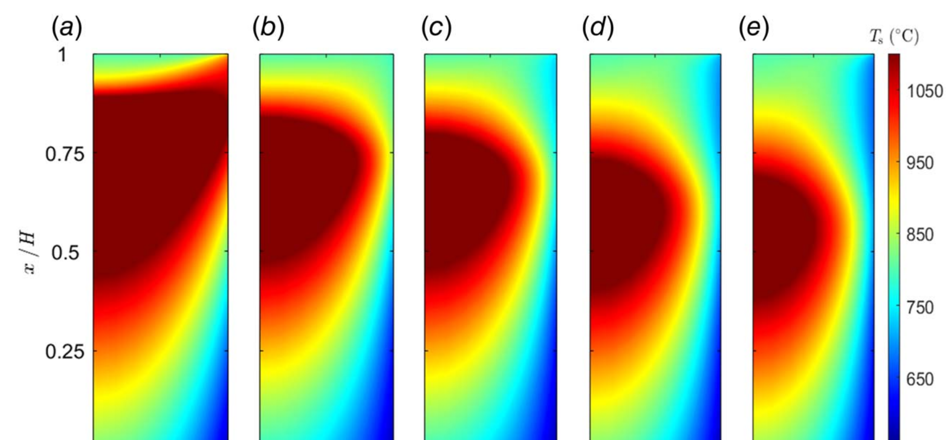

Fig. 10 Particle bed temperature contours for (*a*) $C_{HX} = 0.5$, (*b*) $C_{HX} = 2.5$, (*c*) $C_{HX} = 4.5$, (*d*) $C_{HX} = 6.5$, and (e) $C_{HX} = 8.5$

temperature >1050 °C can be obtained for $C_{HX} = 0.5$, showing that in conjunction with high-temperature resistant material, such an OR/HX can be used for ultra-high-temperature power cycles. The Y_{O_2} profiles in Fig. 11(*c*) are analogous in trend and magnitude, and the α profiles in Fig. 11(*d*) follow the general trend of decreasing α for increasing C_{HX}, where a higher α is due to a larger high-temperature zone. The $\dot{Q}$ values are presented in Table 10, where $\dot{Q}$ ranges from 17.90 kW to 16.02 kW for $C_{HX} = 0.5$–8.5. The decrease in $\dot{Q}$ with higher C_{HX} can be attributed to the shrink of the high-temperature zone in the particle bed, as observed in Fig. 10.

4.2.3 Effect of Particle Channel Aspect Ratio. To study the effects of the AR_s on the OR/HX performance, the channel height H is kept constant, and w_s is varied to increase AR_s from 16 to 48. It is also noted that for the present cases, the superficial velocity

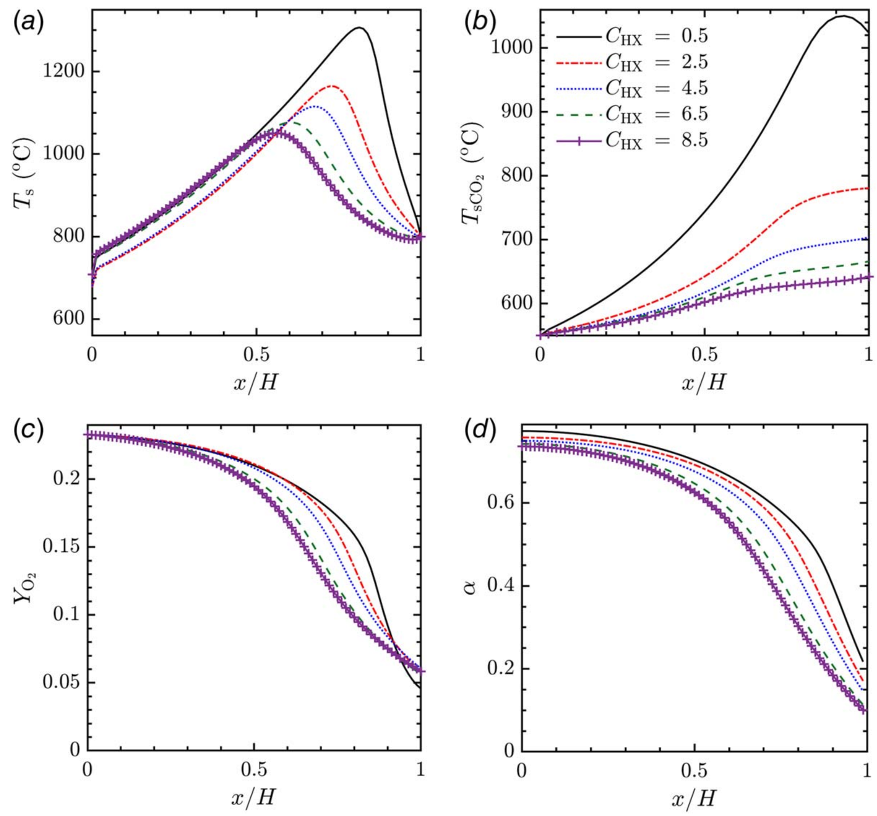

Fig. 11 (*a*) Particle bed temperature, (*b*) sCO_2 temperature, (*c*) O_2 mass fraction, and (*d*) extent of reaction profiles along the particle flow direction for different C_{HX} ($H = 0.4$ m, $w_s = 25.4 \times 10^{-3}$ m, $w_{sCO_2} = 5 \times 10^{-4}$ m, $t_w = 1 \times 10^{-3}$ m, $\dot{m}_s = 10$ g/s, $\dot{m}_{g,in} = 3$ g/s, $T_{s,in} = 800\,^\circ$C, $T_{g,in} = 550\,^\circ$C, $T_{sCO_2,in} = 550\,^\circ$C, and $d_s = 1 \times 10^{-3}$ m)

of the particle bed is constant, which would translate to higher mass flowrates for lower AR_s. The particle bed temperature contours are shown in Fig. 12, where a very large high-temperature zone is noted for $AR_s = 16$, and it decreases as AR_s increases to 48, which is due to the high mass flowrates of the particle bed in lower AR_s cases and the inability of the sCO_2 channel to extract the heat generated due to limited heat capacitance. As noted in Sec. 4.2.2, high temperatures near the dividing walls should be avoided to enhance the life of the dividing wall; hence, lower AR_s cases may not be appropriate for high lifecycle operation.

The streamwise averaged temperature profiles of the particle bed are shown in Fig. 13(*a*). The high-temperature zones are present between $0.5 < x/H < 0.9$. Following the particle bed temperature profile, the sCO_2 temperature profiles show analogous behavior, where sCO_2 outlet temperature >750 °C is observed for $AR_s = 16$. The sCO_2 outlet temperature then decreases as AR_s increases. The Y_{O_2} and α profiles follow a similar distribution in Figs. 13(*c*) and 13(*d*) due to the strong species transport-reaction coupling. The $\dot{Q}$ values are presented in Table 10, where $\dot{Q}$ ranges from 24.54 kW to 7.76 kW as AR_s increases from 16 to 48.

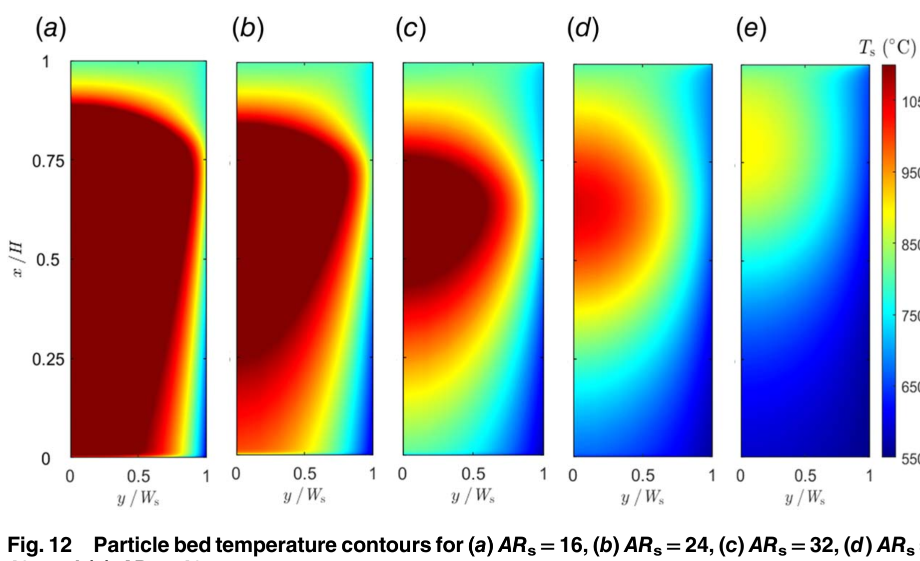

Fig. 12 Particle bed temperature contours for (*a*) $AR_s = 16$, (*b*) $AR_s = 24$, (*c*) $AR_s = 32$, (*d*) $AR_s = 40$, and (*e*) $AR_s = 48$

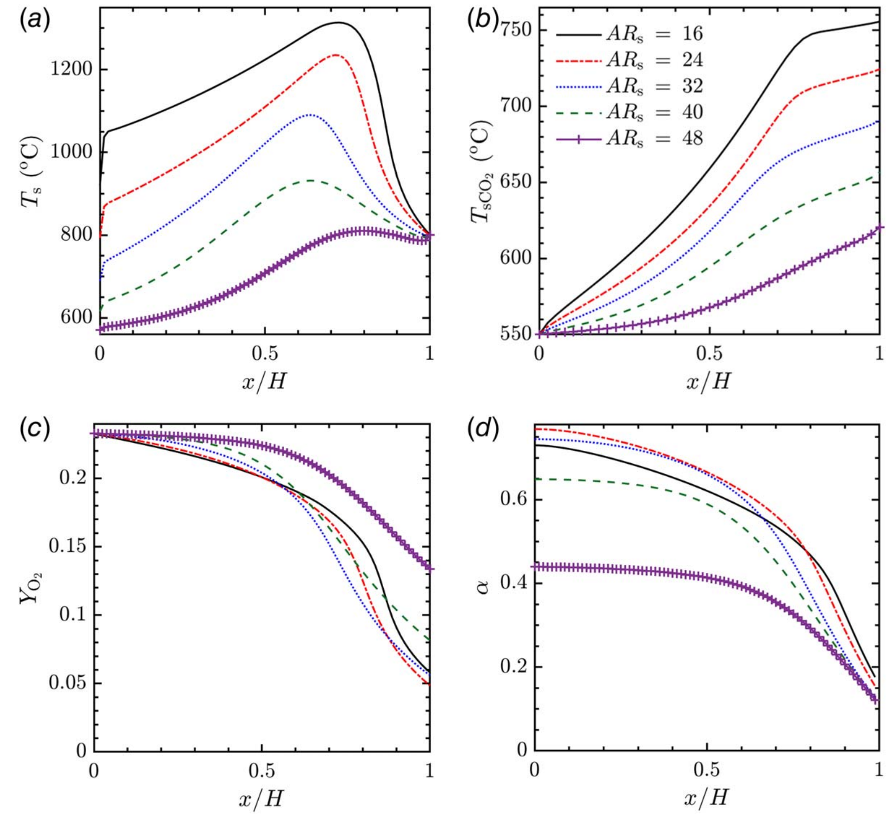

Fig. 13 (*a*) Particle bed temperature, (*b*) sCO_2 temperature, (*c*) O_2 mass fraction, and (*d*) extent of reaction profiles along the particle flow direction for different AR_s ($H = 0.4$ m, $w_{sCO_2} = 5 \times 10^{-4}$ m, $t_w = 1 \times 10^{-3}$ m, $\dot{m}_s = 10$ g/s, $\dot{m}_{g,in} = 3$ g/s, $\dot{m}_{sCO_2} = 38$ g/s, $T_{s,in} = 800\,^\circ$C, $T_{g,in} = 550\,^\circ$C, $T_{sCO_2,in} = 550\,^\circ$C, and $d_s = 1 \times 10^{-3}$ m)

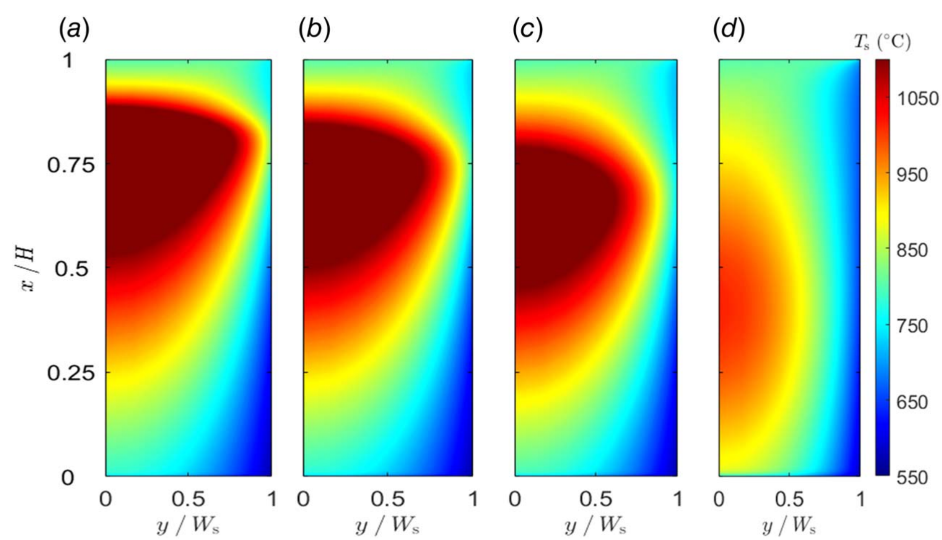

Fig. 14 Particle bed temperature contours for (*a*) $d_s = 0.5$ mm, (*b*) $d_s = 0.75$ mm, (*c*) $d_s = 1$ mm, and (*d*) $d_s = 2$ mm

4.2.4 Effect of Average Particle Diameter. The effect of d_s (varied from 0.5 mm to 2 mm) on the OR/HX performance is studied in this section. The particle bed temperature contours are shown in Fig. 14, where the high-temperature zone is analogous to the baseline case (Table 7) from $x/H = 0.3$–0.8 for $d_s = 0.5$–1 mm and starts dissipating as d_s increases to 2 mm. This is due to d_s influencing the reaction rate, with smaller d_s leading to higher reaction rates.

The streamwise averaged temperature profiles of the particle bed are shown in Fig. 15(*a*), where a gradual decrease in the maximum temperature from 1200 °C to 850 °C is observed as d_s increases. Similarly, for the sCO_2 temperature profiles in Fig. 15(*b*), higher sCO_2 outlet temperatures are obtained for 0.5 mm < d_s < 1 mm compared to $d_s = 2$ mm. This reduction in outlet temperature for $d_s = 2$ mm can also be attributed to the higher near-wall resistance according to Eq. (27). For the Y_{O_2} and α profiles in Figs. 15(*c*)

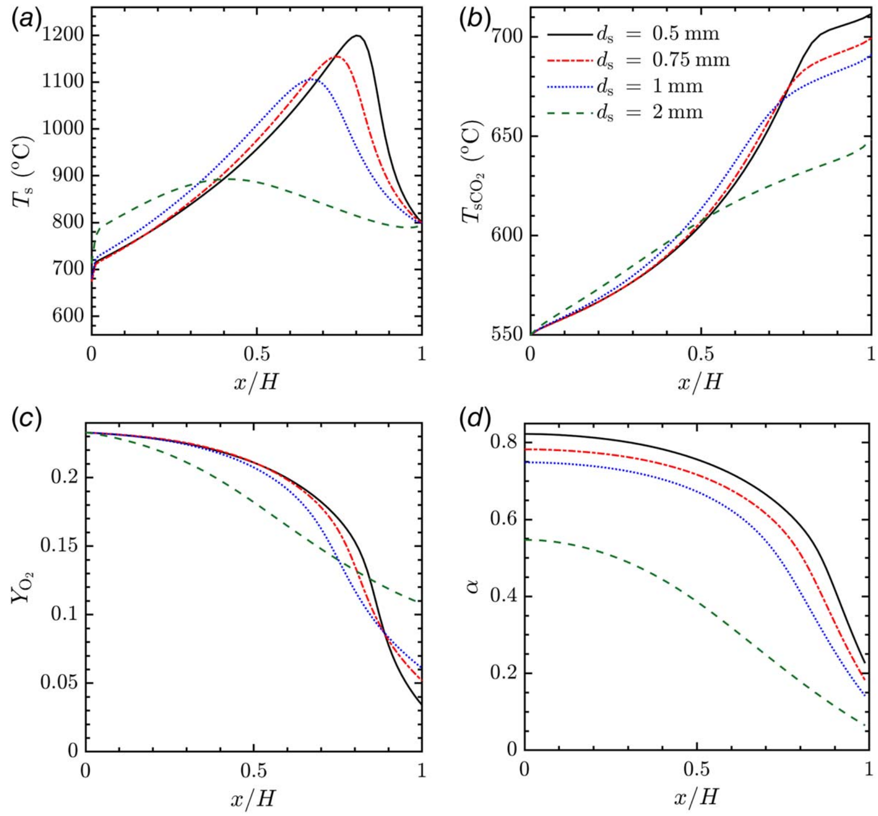

Fig. 15 (*a*) Particle bed temperature, (*b*) sCO_2 temperature, (*c*) O_2 mass fraction, and (*d*) extent of oxidation profiles along the particle flow direction for different d_s ($H = 0.4$ m, $w_s = 25.4 \times 10^{-3}$ m, $w_{sCO_2} = 5 \times 10^{-4}$ m, $t_w = 1 \times 10^{-3}$ m, $\dot{m}_s = 10$ g/s, $\dot{m}_{g,in} = 3$ g/s, $\dot{m}_{sCO_2} = 38$ g/s, $T_{s,in} = 800\,°C$, $T_{g,in} = 550\,°C$, and $T_{sCO_2,in} = 550\,°C$)

and 15(*d*), the trends suggest that lower d_s favors higher α, which causes higher consumption of O_2. The $\dot{Q}$ values ranging from 18.97 kW to 11.26 kW are presented in Table 10, showing the high influence of d_s on the HX capacity.

5 Conclusions

In this paper, a 2D heat and mass transfer model is developed to investigate the thermochemical heat extraction from a moving packed-bed oxidation reactor (OR) to an indirect shell-and-plate particle-to-sCO_2 HX (OR/HX). The continuum model couples the mass, momentum, species, energy transport, and reaction kinetics in corresponding zones of the particle bed, gas mixture, dividing wall, and sCO_2 working fluid. The coupled model predicts the distributions of temperature, O_2 concentration, and extent of oxidation in the OR as well as the sCO_2 temperature and the total heat transfer rate in the HX. The following major conclusions are drawn from this study:

- For the OR/HX with baseline parameters, a maximum particle bed temperature of 1100 °C was observed, the maximum extent of oxidation at the particle bed outlet was ∼75%, and the sCO_2 outlet temperature was 692 °C compared to 595 °C for the sensible heat HX with the same baseline design parameters. The total heat transfer rate for the OR/HX was 16.71 kW compared to 4.62 kW for sensible heat HX for a 1.0 m^2 heat transfer area.
- Since the OR/HX baseline conditions do not correspond to optimal sensible heat HX operation, optimal parameters were adopted from Ref. [39], and a total heat transfer rate of 10–20 kW was obtained for narrow-channel sensible heat HXs with different particle bed inlet temperatures between 800 °C and 1000 °C.
- A detailed parametric study was also conducted to evaluate the effects of major design and operating parameters, including particle bed, gas mixture, and sCO_2 mass flowrates, particle bed channel width and height, and average particle diameter on the distributions of particle bed temperature, O_2 mass fraction, reaction extent, sCO_2 temperature, and total heat transfer rate of the OR/HX.

Although the integrated OR/HX shows better heat transfer performance compared to sensible heat HX, the design and operation of OR/HX can be much more challenging considering the coupling between heat transfer and chemical reactions. There is also additional energy input necessary to reduce the particles in a reduction reactor that could be integrated with a solar receiver (analogous to sensible heating of inert particles in the charging step). Further insights into high-temperature materials that can be used to manufacture such OR/HX also need to be the focus from a material development aspect. Future work on designing and experimental testing of such integrated OR/HX is necessary to validate the model and demonstrate the technical feasibility.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

d = particle diameter (m)
h = heat transfer coefficient (W/m^2K)
t = thickness (m)
w = channel width (m)
A = specific surface area (m^{-1})
D = diffusivity (kg/ms)
E = internal energy (kJ/kg)
H = channel height (m)
T = temperature (°C)
$\dot{Q}$ = heat transfer rate (W)
c_p = specific heat capacity (J/kgK)
C_{bed} = particle bed capacitance ratio
C_{HX} = HX capacitance ratio
ARs = aspect ratio of particle channel

Greek Symbols

ε = porosity
λ = thermal conductivity (W/m/K)
μ = viscosity (kg/m/s)
$\vec{v}$ = velocity (m/s)
ρ = density (kg/m^3)
p = pressure (Pa)
Y = mass fraction

Subscripts

bed = particle bed
g = gas mixture phase
ox = oxidation
p = particle
s = particle bed phase
sCO_2 = supercritical CO_2
SH = sensible heat
TC = thermochemical
w = wall

Abbreviations

HTF = heat transfer fluid
HX = heat exchanger
LHS = latent heat storage
MPBHX = moving packed-bed heat exchanger
OR = oxidation reactor
SHS = sensible heat storage
TCES = thermochemical energy storage
TES = thermal energy storage
WF = working fluid

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